May 10, 2026

CURRICULUM VITAE – Abridged Version

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Citizenship: U.S.

Positions Held

Staff Mathematician, Applied and Computational Mathematics Division, NIST	2023–Now
Instructor, North Carolina School of Science and Mathematics	2023
Flatiron Research Fellow, Center for Computational Biology, Flatiron Institute	2018–2022
Supervisor: Prof. Michael Shelley	

Visiting Positions

Visitor, Courant Institute of Mathematical Sciences, New York University	2019–2020
Visiting Scientist, Department of Physics, Brown University	2018–2019

Education

Ph.D. Applied Mathematics, Brown University	2018
Advisors: Profs. Thomas Powers & Robert Pelcovits	
Thesis: "Geometry and Mechanics of Self-Assembling Colloidal Membranes"	
Sc.M. Engineering, Brown University	2017
Sc.M. Applied Mathematics, Brown University	2013
B.S. Physics, Georgia Institute of Technology	2012
B.S. Applied Mathematics, Georgia Institute of Technology	2012
Certificate in I/O Psychology, Georgia Institute of Technology	2012

Awards and Honors

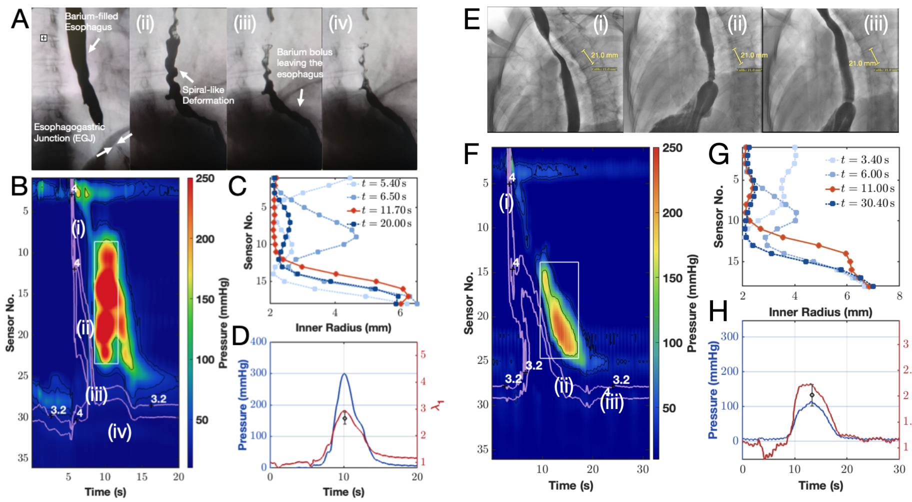
Sigma Xi, Brown University Chapter	2018
Outstanding Teaching Assistant, Brown University	2014
School of Mathematics MCTP Scholarship, Georgia Institute of Technology	2012, 2011
Hitohiro Fukuyo Memorial Award, Georgia Institute of Technology	2011

Research Interests

Applied mathematics: analytical & computational modeling, PDE, asymptotics, geometry
Theory of soft matter: membranes, fluid interfaces, colloids, liquid crystals, active matter
Biomechanical engineering: complex fluids, elasticity, self-assembly, bio-inspired materials

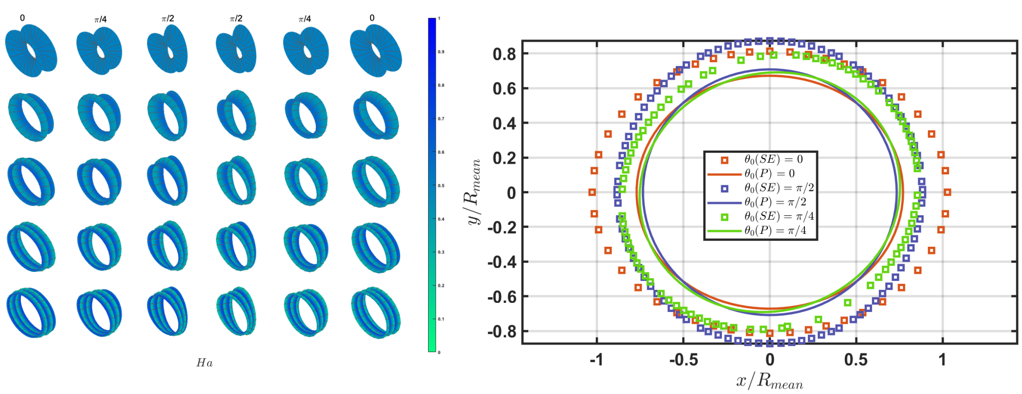
Projects in preparation (drafts available upon request)

- X. Liu, L. L. Jia, M. M. Alikhan, W. Kou, D. A. Carlson, P. J. Kahrilas, J. E. Pandolfino, N. A. Patankar. "Mechanistic analysis of esophageal corkscrewing driven by active contraction." To be submitted Summer 2026.



Abstract: Corkscrew esophagus is a pathological condition in which the esophageal wall undergoes helical twisting or localized constrictions in response to strong muscular contractions. To unravel the mechanical origins of this phenomenon, we build a finite-deformation minimal model of the esophageal wall that incorporates an active thin-shell outer layer to capture the effect of contracting muscle fibers. We apply this model to measure muscle contractility in vivo using high-resolution manometry, and the corkscrew is predicted to emerge as an elastic instability when critical values of contractility are achieved. Analysis of the model and clinical data reveal plausible physiological changes in the esophageal muscle and matrix that precipitate instability including longitudinal muscle impairment and material softening due to repeated straining. Our framework links clinical observations to first-principles mechanics and provides quantitative targets for diagnostic thresholds and therapies.

- X. Liu, L. L. Jia. "A Plateau problem for membranes." To be submitted 2026.



Abstract: We use perturbation theory and numerical computation to solve a Plateau-type problem for fluid membranes with fixed area and resistance to bending. In our version of the problem, we are given two closed curves, which may not necessarily be symmetric, planar, or concentric, and we determine the shape of the membrane that minimizes the Canham–Helfrich bending energy. Since the perturbed surfaces are not necessarily functions of the axial extension, a special coordinate frame is devised to parameterize these highly deformed shapes. The permissible surfaces are found to be generalizations of minimal surfaces such as the catenoid that characterize the shapes of soap films; these surfaces can also display hallmarks of confined elastic surfaces such as buckling and wrinkling. The connection between these two systems is made precise by a one-to-one correspondence between stability eigenvalues. Our mathematical description provides insight into the forces and torques needed to stabilize cellular membranes and other elastic interfaces as well as methods that could be used to measure the elasticity of these objects.

Publications

8. G. R. Chickering, L. L. Jia, M. DiSalvo, M. A. Catterton, P. N. Patrone, E. M. Darling, G. A. Cooksey. "Estimating single-cell elastic modulus in a serial microfluidic cytometer from time-of-flight and fluorescence signals analysis." Lab on a Chip (2026). Advance article.
7. R. Adkins, J. Robaszewski, S. Shin, F. Brauns, L. L. Jia, A. Khanra, P. Sharma, R. A. Pelcovits, T. R. Powers, Z. Dogic. "Topology and kinetic pathways of colloidosome assembly and disassembly." Proceedings of the National Academy of Sciences of the USA **112**(36) (2025) e2427024122.
6. L. L. Jia, W. T. M. Irvine, M. J. Shelley. "Incompressible active phases at an interface. Part 1. Formulation and axisymmetric odd flows." Journal of Fluid Mechanics **951** (2022) A36.
5. L. L. Jia, M. J. Shelley. "The role of monolayer viscosity in Langmuir film hole closure dynamics." Journal of Fluid Mechanics **948** (2022) A1.
4. A. Khanra, L. L. Jia, N. P. Mitchell, A. Balchunas, R. A. Pelcovits, T. R. Powers, Z. Dogic, P. Sharma. "Controlling the shape and topology of two-component colloidal membranes." Proceedings of the National Academy of Sciences of the USA **119**(32) (2022) e2204453119.
3. L. L. Jia, S. Pei, R. A. Pelcovits, T. R. Powers. "Axisymmetric membranes under external force: buckling, minimal surfaces, and tethers." Soft Matter **17** (2021) 7268. Selected as Front Cover.
2. A. Balchunas, L. L. Jia, M. J. Zakhary, J. Robaszewski, T. Gibaud, Z. Dogic, R. A. Pelcovits, T. R. Powers. "Force-induced formation of twisted chiral ribbons." Physical Review Letters **125** (2020) 018002.
1. L. L. Jia, M. J. Zakhary, Z. Dogic, R. A. Pelcovits, T. R. Powers. "Chiral edge fluctuations of colloidal membranes." Physical Review E **95** (2017) 060701(R).

In the News

1. L. L. Jia and X. Liu. "Mathematical Models Can Help Us Understand – and Possibly Treat – Complex Diseases." NIST Taking Measure Blog (2025).

Recent Presentations (from the last 5 years)

- Contributed talk, APS Division of Fluid Dynamics Meeting (November 2025)
- Invited talk, SIAM/CAIMS Annual Meeting (July 2025)
- Poster, NIST ITL Science Day (December 2024)
- Contributed talk, APS Division of Fluid Dynamics Meeting (November 2024)
- Poster, International Soft Matter Conference (July 2024)
- Contributed talk, APS Division of Fluid Dynamics Meeting (November 2023)
- Seminar, National Institute of Standards and Technology (February 2023)

- Seminar, University of Alabama at Birmingham (February 2023)
- Seminar, University of Pennsylvania (January 2023)
- Seminar, National Security Agency (October 2022)
- Seminar, Flatiron Institute (October 2022)
- Seminar, Manhattan College (August 2022)
- Seminar, High Point University (July 2022)
- Seminar, Soft, Living, Active, and Adaptive Matter Seminar Series (June 2022)
- Poster, Mechanics of Life Workshop (May 2022)
- Seminar, The Cooper Union (April 2022)
- Seminar, Northwestern University (February 2022)
- Contributed talk, APS Division of Fluid Dynamics Meeting (November 2021)
- Seminar, Flatiron Institute (September 2021)

Teaching Experience

- Lecturer, NIST ACMD Reading Group (January 2024–June 2024). Topics included the calculus of variations, differential geometry, and theoretical mechanics.
- Instructor, North Carolina School of Science and Mathematics (Spring 2023). Taught biomedical engineering.
- Instructor, Division of Applied Mathematics, Brown University (Summer 2017). Taught introductory scientific computing.
- Teaching Consultant, Sheridan Center for Teaching and Learning, Brown University (Fall 2017–Spring 2018). Obtained Sheridan Center Reflective Teaching, Course Design, and Teaching Consultant Certificates.
- Graduate Teaching Assistant, Division of Applied Mathematics, Brown University (Fall 2013–Spring 2015). Taught methods of applied mathematics.
- Math Resource Center Tutor, Brown University (Fall 2012–Spring 2018). Tutored undergraduate mathematics.
- Content Instructor and Peer Mentor, New Scientist Program Catalyst, Brown University (Summer 2014). Taught complex numbers and linear stability theory.
- Undergraduate Teaching Assistant, School of Physics, Georgia Institute of Technology (Spring 2009–Spring 2012). Taught introductory mechanics and electromagnetism.
- Undergraduate Teaching Assistant, School of Mathematics, Georgia Institute of Technology (Fall 2011–Spring 2012). Taught calculus and linear algebra.
- Undergraduate Grader, School of Mathematics, Georgia Institute of Technology (Fall 2010–Spring 2012). Graded for partial differential equations, abstract algebra, and real analysis courses.
- I regularly volunteer as a freelance tutor 5 to 10 hours/week. I teach K12 and college math and science.

Students Trained

- Henry Nonnemaker, SURF student at NIST (Summer 2026). Henry is an undergraduate student at Case Western Reserve University.
- Xinyi Liu, Guest Researcher at NIST (Spring 2024–present). Co-advised with N. A. Patankar. Xinyi is a Ph.D. student in the Department of Engineering Sciences and Applied Mathematics at Northwestern University.
- Sydney Lee, SURF student at NIST (Summer 2024). Sydney is an undergraduate student at Texas A&M University.
- Steven Pei, UTRA student at Brown University (Fall 2016–Fall 2018). Co-advised with T. R. Powers. Steven studied Physics and is currently a software engineer at Google.

Recent Service (from the last 5 years)

- Co-chair (w/ Z. Grey), NIST ACMD Seminar Series (Summer 2024–present)
- Events Committee, Association of NIST Asian-Pacific Americans (Fall 2023–present) & Secretary, Association of NIST Asian-Pacific Americans (Fall 2024–present)
- Georgia Tech Alumni Association Mentor (Fall 2024–present)
- Reviewer, NIST Small Business Innovation Research grant proposals (Spring 2024)
- Member, Simons Foundation AAPI Employee Resource Group (Winter 2021–Fall 2022)
- Professional Development Committee, Flatiron Institute (Winter 2021–Fall 2022)
- Referee, Journal of Fluid Mechanics (×5)

Personal

- Languages: English (native), Mandarin Chinese (fluent), Japanese (conversational)
- Memberships: APS (DFD, DSOFT, DBIO), SIAM

References

Mr. Joseph Browne (teaching reference)
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Prof. Zvonimir Dogic (frequent collaborator)
Dept. of Physics & Dept. of Biomolecular Science and Engineering, UCSB
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Prof. Robert Pelcovits (thesis advisor)
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Prof. Thomas Powers (thesis advisor)
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Prof. Michael Shelley (postdoctoral supervisor)
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